

GenAI-Powered Customer Support Quality Auditor

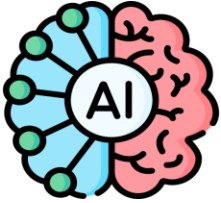
Submitted to:



Group1 – Team 2

**Under the Guidance of
mentor: NISHYANTH**

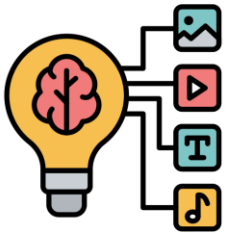
ABSTRACT



The **GenAI-Powered Customer Support Quality Auditor** is an intelligent automation platform that revolutionizes how enterprises evaluate and improve customer support quality.



Using **Google Gemini LLM**, **advanced NLP**, and **speech-to-text capabilities**, the system automatically analyzes chat logs and **call recordings** to assess key **quality metrics** — including empathy, professionalism, compliance, resolution effectiveness, and tone



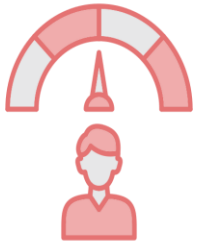
Powered by a LangChain + Qdrant **RAG pipeline**, the platform delivers context-aware auditing by cross-referencing conversations against **company policies** and SOPs in **real time**. It **detects compliance violations instantly**, assigns objective quality scores, and provides personalized improvement suggestions to agents.

By replacing manual QA processes with **AI-driven intelligence**, this platform significantly **reduces audit time**, ensures consistent **quality standards**, improves agent **performance**, and enhances overall customer experience.

INTRODUCTION



In today's competitive landscape, delivering exceptional customer support is no longer optional — it's a key differentiator for businesses. However, **traditional quality auditing processes** remain largely manual, **inconsistent**, **time-consuming**, and subjective.



Supervisor Struggles:

Reviewing thousands of interactions manually.
Subjective and **inconsistent scoring** across auditors.
Delayed feedback leading to **performance stagnation**.



GOAL

Revolutionize **QA using Generative AI for real-time**, objective evaluation.





Problem Statement



Manual Oversight:

Human reviewers can only audit a fraction of calls, **leaving 98%** of data unmonitored.

Regulatory Risk:

Failure to ensure **100% script compliance leads** to legal liabilities.

Inconsistent Quality:

Subjective grading makes it **impossible to standardize agent coaching**.

High Operational Cost:

Large QA teams are expensive and difficult to scale linearly.

Delayed Intelligence:

Critical issues are often discovered **weeks after they occur**.

Objectives



Implement **automated transcription** with high-fidelity **speaker diarization**.

Develop a **RAG pipeline** for context-aware **compliance auditing against company SOPs**.

Ensure **100% transparency** and scalability for enterprise-grade support centers.

Build a **GenAI-based scoring engine** for Empathy, Professionalism, and Resolution.

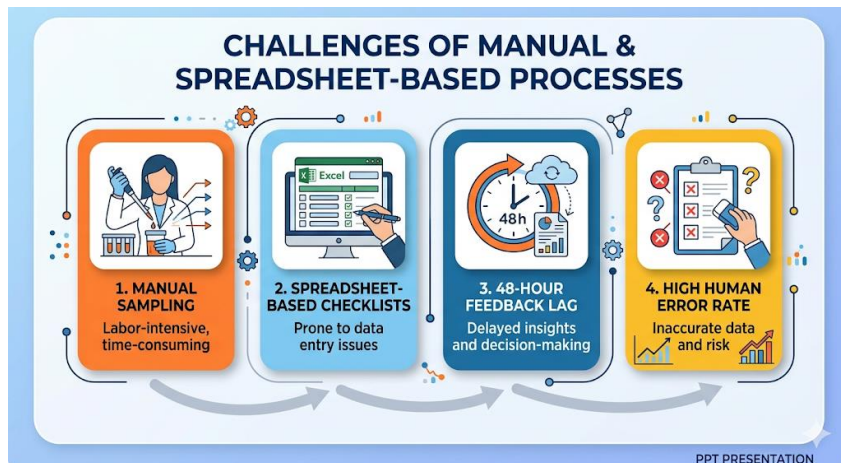
Provide a **real-time analytics** hub for supervisors with automated "Critical" alerting.

Existing vs. Proposed System



Existing System

Manual sampling, **spreadsheet**-based checklists, **48-hour feedback lag**, high **human error** rate.



Proposed System (AuditAI)

100% Audit Coverage: Every call is graded instantly.

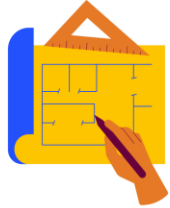
Real-Time Alerts: Immediate flags for compliance breaches.
Objective

Scoring: Metrics based on standardized LLM reasoning.

Actionable Feedback: Suggestive coaching markers generated for every agent.



AuditAI
GENAI SUPPORT QUALITY AUDITOR



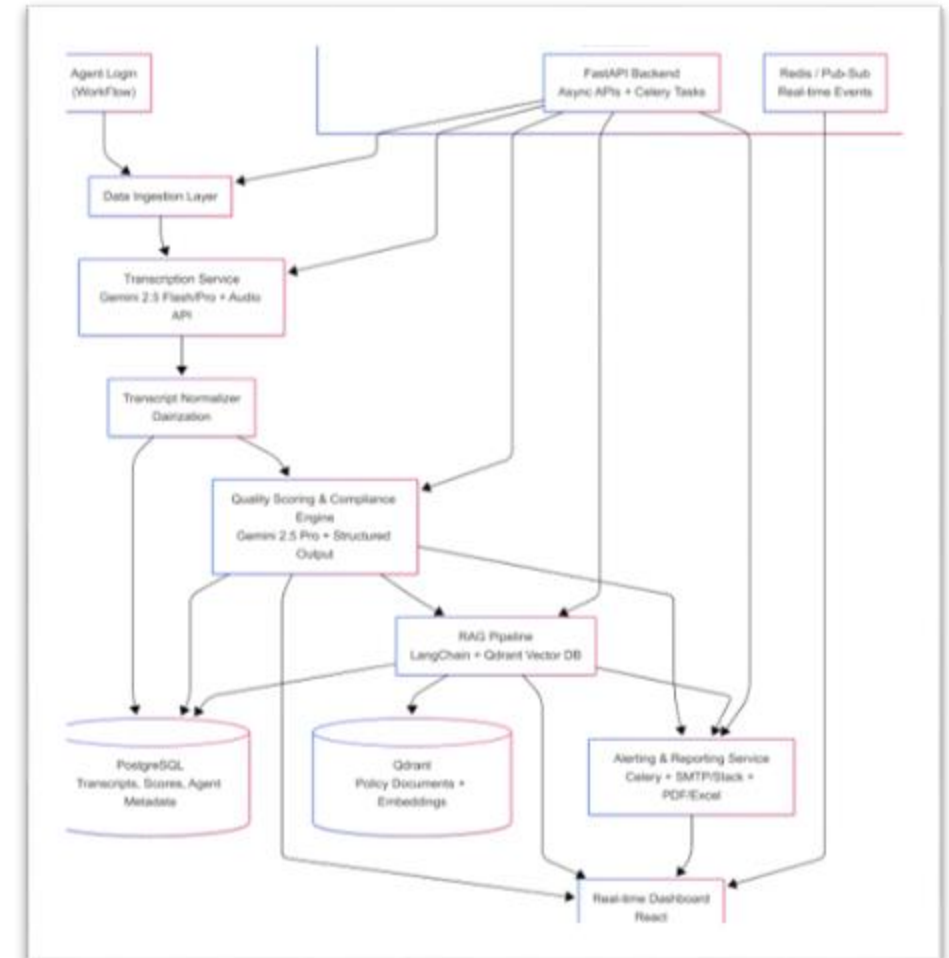
System Architecture

Data Tier: **PostgreSQL** (Supabase) + Qdrant (**Vector Knowledge Base**). AI Tier: **Gemini 1.5** Flash + LangChain (**RAG Orchestrator**).

Logic Tier: **FastAPI** (Asynchronous Python) + JWT Security.

Presentation Tier: **React** Dashboard (TailwindCSS + Lucide Icons).

Workflow: Data Ingestion → **AI Diarization** → **RAG Retrieval** → **Compliance Auditing** → **Supervisor Alerting**.



[Click to View](#)

Technologies Used



Google Gemini (AI Studio)

Provides **deep reasoning** and **native transcription**.

LangChain

Orchestrates the complex **RAG** and **auditing pipelines**.

Supabase & Qdrant Vector DB:

Stores and **retrieves company policies** with **semantic precision**.

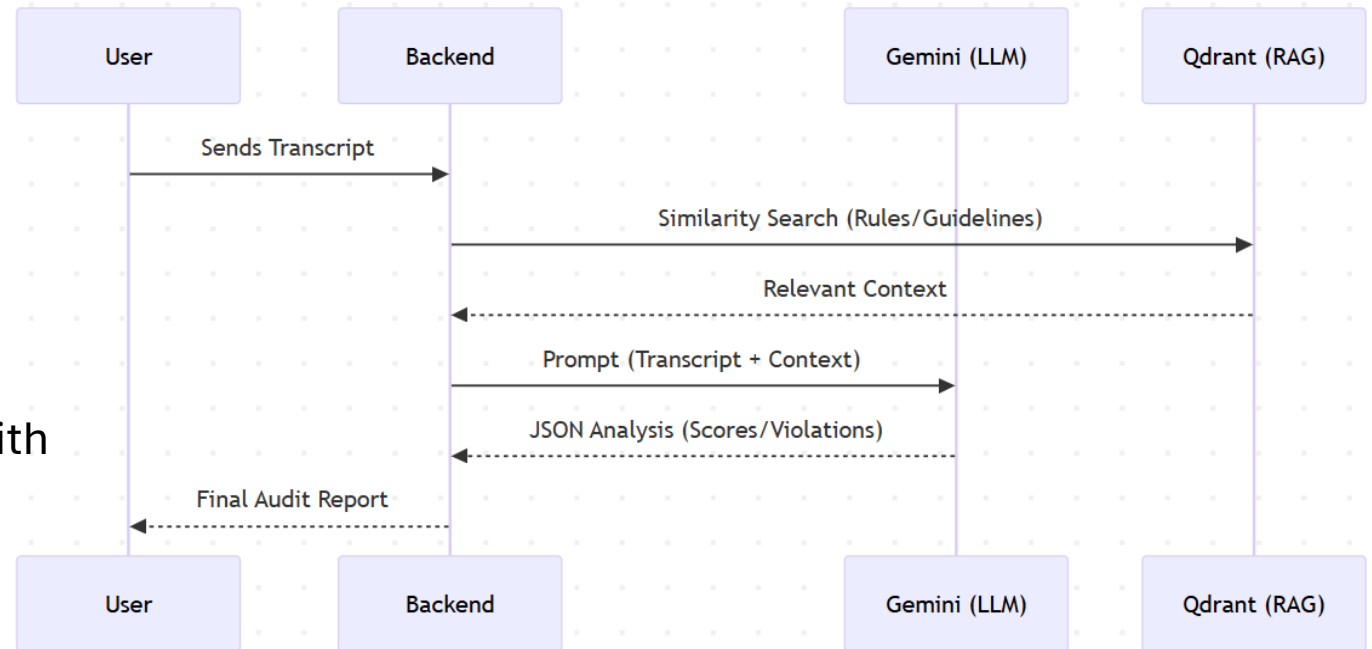
PostgreSQL(Supabase)

FastAPI

Powers the high-concurrency backend with **thread-safe connection pooling**.

React

Delivers a **premium**, responsive **real-time analytics dashboard**.



[View](#) Real-time Work-Flow

Results & Output



Deployment-Frontend

Vercel :

[Build and deploy on the AI Cloud](#)

Docker

Containers share the OS → faster and use less memory.

GenAI-Audit



Deployment-Backend

Railway:

[Railway is a full-stack cloud for deploying web apps, servers](#)



Deployment-Database

[GenAI-Audit](#) - <https://ai-powered-customer-support-quality.vercel.app/>



You Can Access App Here

Advantages

Scalability:

Monitor thousands of concurrent threads without increasing audit headcount.

Compliance Security:

100% verification against legal and internal scripts.

Data-Driven Training:

Precise **coaching suggestions** for every agent interaction.

Operational Speed:

Feedback loops reduced from days to under 60 seconds.

Cost Efficiency:

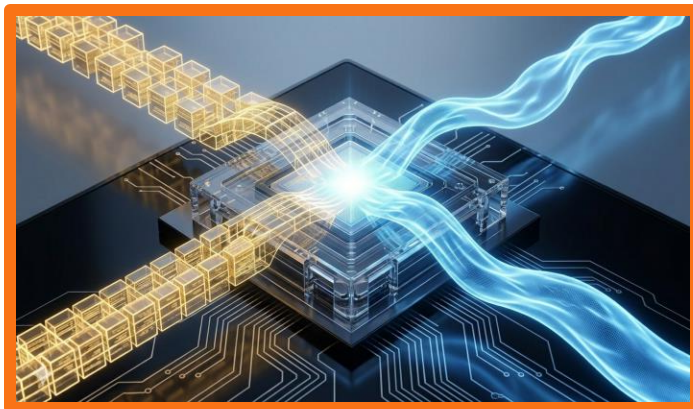
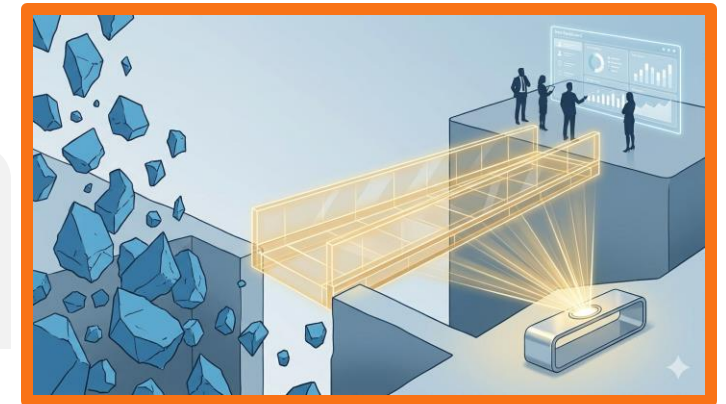
Significant **reduction** in manual QA expenditure.

Conclusion



AuditAI represents a paradigm shift in Quality Assurance.

By combining Gemini's Reasoning with RAG's Context, it provides a scalable, objective solution for enterprise compliance and quality growth.



The system successfully bridges the gap between raw conversational data and strategic management insights.

Q&A



Thank You!

Questions?

